Modeling asymmetric Thermal Performance Curves for proportion data

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Abstract

something here

1 Introduction

Ambient temperature impacts a variety of functional traits, such as fecundity, development, or survival, especially in ectotherms REF. Quantifying the impact of temperature on traits is usually achieved through empirical, laboratory based experimental studies. One approach is to hold organisms at an array of fixed temperatures and measure focal traits at each of these temperatures. The pattern of the trait values across temperature are then often quantified using Thermal Performance Curve (TPCs, Huey and Stevenson 1979; Schoolfield, Sharpe, and Magnuson 1981; Amarasekare and Savage 2012). These TPCs are usually functions that are non-negative (as measured traits are typically non-negative). They are often hump- or u-shaped, and may be either symmetric or asymmetric (see Amarasekare and Savage (2012) and Gajewski et al. (2021) for a range of examples).

Often the observed trait measurements (or the average of these) may reasonably be assumed to be roughly normally distributed at a particular temperature, with the mean at that temperature described by a TPC. For example, across moderate temperatures mosquitoes often lay many eggs at once, so these counts (or the average across times/individuals) may be approximately normal. In these cases, standard fitting approaches including least squares may be used to fit the TPCs to data (Padfield, O'Sullivan, and Pawar 2021). Further, as long as the data may reasonably be assumed to be non-negative and likely unbounded from above, the general approach may still work. For example, if data are counts of events, such as eggs laid, then the

standard TPC can be used, but linked to other distributions, such as Poisson or a truncated normal, in order to better describe the variability of the data around the assumed TPC.

However, this approach is not ideal for data which are bounded. Specifically, proportion data, where the number of "successes" within a given number of trials are recorded, are poorly captured by the standard TPC framework. This is because two conditions must hold in order for these types of data to be approximated by a normal distribution. First the probability of success must not be "too close" to either 1 or zero. This is problematic in the current TPC framework because we often specifically want the TPC to be close to these limits at particular temperatures. Further, the number of trials must be sufficiently large, with the sample size needing to be even higher the closer that the success probability is to either zero or 1. If these conditions do not hold, the point estimate of the mean may be reasonable, but any uncertainty bounds will be problematic. For example, if the sample size is small, the error may be poorly estimated by the normal distribution. Additionally, when the uncertainty is problematic in the TPC where probabilities are near 1 or 0, the estimated ranges may not correspond to appropriate probabilities (i.e., they include values greater than 1 or less than 0). Thus, while non-linear least squares (NLLS) and other similar fitting approaches can be used to obtain point estimates of TPC parameters, and estimate the resulting mean response, the uncertainty bounds for any parameters and the associated TPC are not meaningful. In addition to the complications of fitting needs, TPCs, in practice, are often constrained by the number of measurements taken in an experiment, along both axes, meaning we wish to minimize the number of parameters.

An alternative approach is to fit the binomial success data directly using generalized linear models (GLMs). In this approach, the data are recorded as the number of successes with the corresponding number of trials instead of as a proportion. A "link" between the modeled probability, formulated in terms of a linear model, and these binomial outcomes is assumed. This approach is flexible and straightforward; however, because of the requirement that the underlying model be linear in the parameters (e.g., lines, polynomial terms or similar), a standard GLM framework cannot capture the asymmetric patterns common in thermal ecology.

In this short note we introduce a simple extension to the standard GLM formulation of TPCs for binomial data using a simple non-linear predictor function. This predictor function is highly flexible, allowing a range

of shapes (concave up or down, left or right skewed), has only 3 parameters to estimate, and is appropriate for use with a binomial modeling approach. First, we introduce the function and describe the parameters and interpretation. Second, we show how to implement it as a TPC to fit trait data in a Bayesian framework using the package `bayesTPC` in R. This allows us to accommodate the non-linear function together with the appropriate link and likelihood. Third, we give examples of fitting the TPC to simulated data, to explore some subtleties inherent in analysis of these kinds of data. Fourth, we use this to fit data across a sampling of arthropod thermal traits. Finally, we discuss the use of the function more generally.

2 Methods

2.1 The Generalized Linear Modeling framework

A Generalized Linear Model (GLM) is a modeling framework that builds on linear regression modelling, a foundational model of statistics. We can write a multiple linear regression (MLR) model mathematically as

$$Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_d X_{id} + \varepsilon_i, \qquad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2).$$

where Y_i is the ith response variable, $X_{i1} \ldots X_{id}$ are the corresponding d predictor variables, the βs are the regression coefficients, and ε is the normally distributed noise with variance σ^2. We can think of an MLR model as having three parts:

1. The ***systematic*** component: the explanatory variables are used to construct a linear predictor $\eta_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_d X_{id}$.
2. The ***random*** component: $Y_1, \ldots, Y_n$ are normally distributed with Y_i having mean μ_i and variance σ^2.
3. The ***link*** between the random and systematic components: $\mu_i = \eta_i, \qquad i = 1, \ldots, n.$

GLMs extend linear models by changing two requirements of the linear model. First, they allow different classes of distributions for the response variables. The most commonly used non-Normal distributions for GLMs are the Binomial (used for binary data) and the Poisson (used for positive counts). Second, they

allow a more general link function:

$$\eta_i = g(\mu_i), \qquad i = 1, \ldots, n,$$

where g is a strictly increasing, twice differentiable function. Each family has a standard, or "canonical" link function. For example, the Poisson uses a log-link.

In this paper, we are interested in binary response data, that is data where the response Y is a binary variable (taking the value 0 or 1). In data on arthropod thermal traits, a common example is whether an egg that is laid successfully becomes an adult. Then a "success" (emergence of the adult) corresponds to the response $Y = 1$, and is $Y = 0$ to a failure. In this case we would use a logistic GLM such that the number of observed successes, n_i, in a fixed number, N, trials is modeled as a Binomial random variable, and assuming a logit link to the linear predictor, that is:

$$n_i \sim \text{Bin}(p_i, N); \quad p_i = \frac{\exp[\eta_i]}{1 + \exp[\eta_i]}; \quad \eta_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_d X_{id}. \tag{1}$$

where again η_i is a linear predictor constructed as above.

In the standard GLM framework the linear predictor has two properties that are important. First, it has to give possible responses across the whole real line – that is when you plug in arbitrary values of predictors it can give values from -∞ to ∞. It is this property that necessitates the link function, which transforms the linear predictor to be within the necessary bounds of the probability model (between 0 and 1 for the binomial). The second property is that the predictor is linear in the *coefficients* of the predictor variables. Thus one can include polynomial terms, for example, because although one is taking powers of the variables, there are no powers (or other functions) of the coefficients in the predictor equation.

2.2 The LINEX TPC

From a theoretical perspective, it is possible to instead use a non-linear predictor – that is, one that fulfills the first requirement but not the second. This is not something that is typically done unless the researcher has a specific reason to desire a different shape than is easy to construct with a linear predictor. In our case, we wish to be able to construct a predictor that is hump/u-shaped (like a quadratic) and asymmetric (unlike

a quadratic) using as few parameters as possible.

One possible solution is to adapt an existing functional form. Linear Exponential (LINEX) loss functions are fairly common as constraints in the economic literature (e.g., Parsian and Kirmani (2002)) and within the statistics literature as an alternative to quadratic loss when estimating model parameters (e.g., Singh, Gupta, and Upadhyay (2002), Li et al. (2007), Doostparast, Ahmadi, and Ahmadi (2013)). The functional form can be modified and used as a TPC in place of a linear predictor to generalize the GLM framework introduced above. We define the non-linear LINEX TPC *predictor* to be

$$f(T) = -e^{(-a(T_{\text{opt}}-T))} + a(T - T_{\text{opt}}) + c \tag{2}$$

where T_{opt} determines the location of the optimum of the function, $c - 1$ is the height of the function at the maximum, and a governs a combination of the curvature (based on the magnitude of a) and skew. The direction of skew is determined by the sign of a: if $a > 0$ it is left skewed and if $a < 0$ it is right skewed (Figure 1, top and middle). The function can be modified slightly to enable a concave up shape, with the value of a still determining the skew,

$$f_{\text{up}}(T) = e^{(-a(T_{\text{opt}}-T))} - a(T - T_{\text{opt}}) + c \tag{3}$$

where the parameters are as for the concave down version, except that value at T_{opt} is now the *minimum* with a value $f_{\text{up}}(T_{\text{opt}}) = c + 1$ (Figure 1, bottom row). Either Equation 2 or Equation 3 can take the place of the linear predictor in the logistic model (Equation 1). We show the inverse logit transform of these equations (corresponding to the probability of success) in the right-hand panels of Figure 1.

We can simulate data from this model to use for testing in a straightforward manner. We define the inverse logit transform of the LINEX function as the probability, p, for the target binomial distribution. We first choose parameters for the LINEX function and the number of trials in an experiment (e.g., the number of eggs). We then draw the number of "successes" (eggs that hatch) using the `rbinom` function in R with the p set as the value of p at the temperature setting, and repeat this across a set number of replicates (Figure 2).

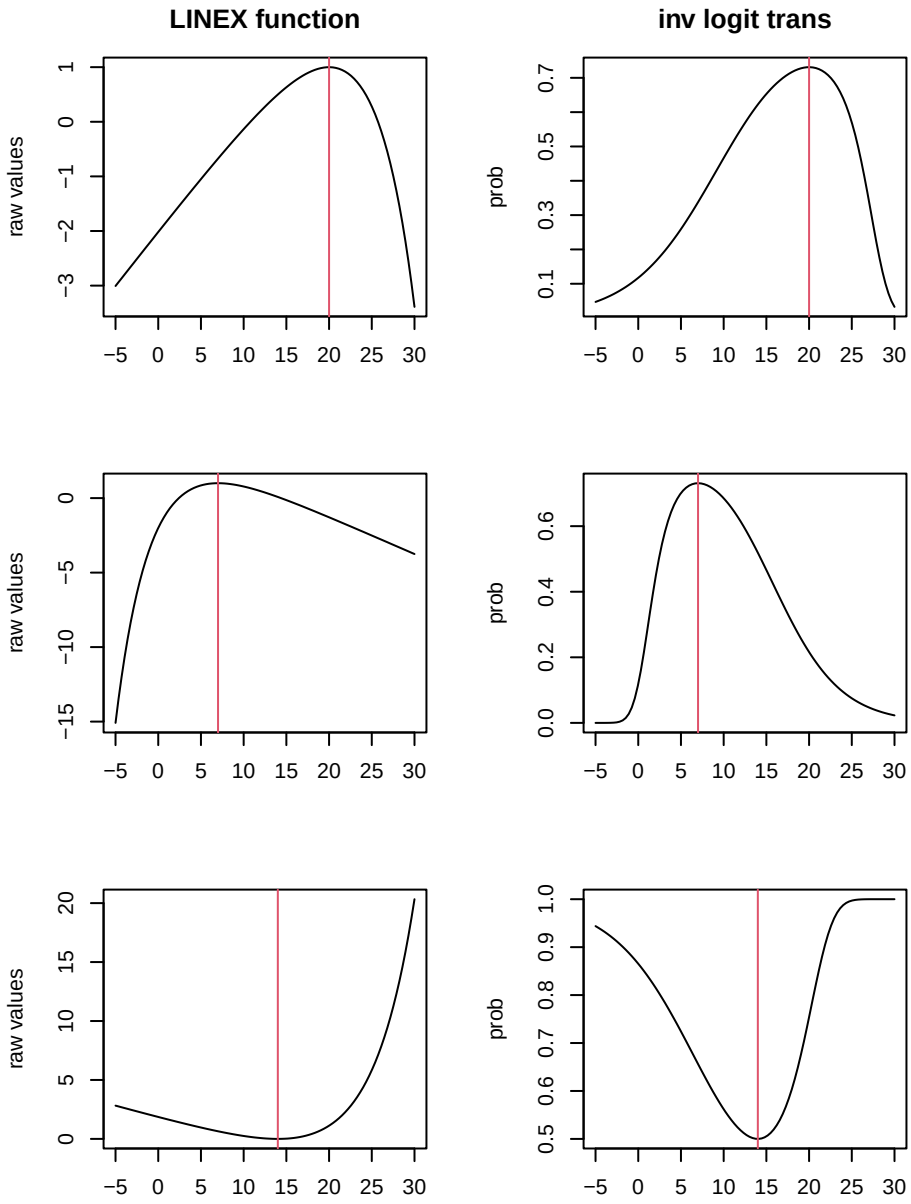


Figure 1: Examples of shapes that can be generated by the LINEX TPC, together with a virtical line indicating the location of the optimum. In each case, the left panel shows the shape of the non-linear predictor and the right panel shows the inverse logit transformed predictor that gives the probability of success. TOP: Left-skewed, $T_{\text{opt}} = 20$, $a = 0.2$, $c = 2$. MIDDLE: Right-skewed, $T_{\text{opt}} = 7$, $a = -0.25$, $c = 2$. BOTTOM: concave-up, $T_{\text{opt}} = 14$, $a = 0.2$, $c = -1$.

We will use these simulated data to test the inference functions in the next section.

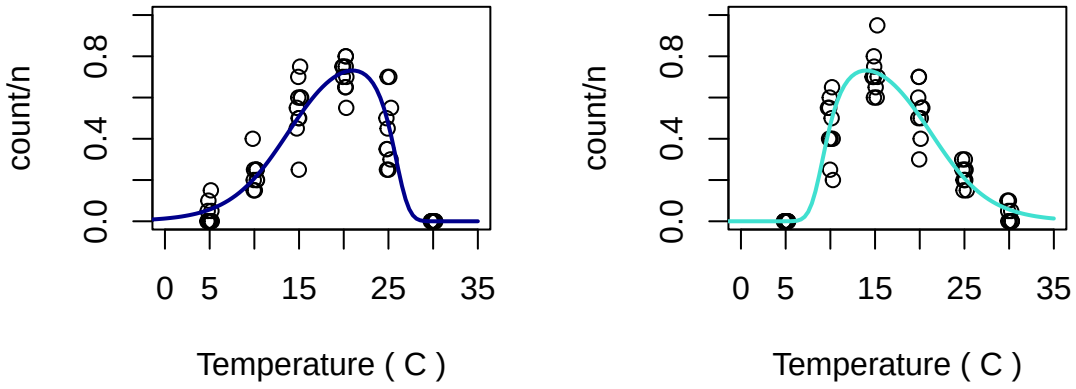


Figure 2: Visualization of simulated data from the LINEX TPC model, together with inverse transform of the assumed functional response (blue lines). Points correspond to 10 virtual replicates, each consisting of $n = 20$ 'trials' with the proportion of successes plotted (counts/n) at each tempearture. Temperature values are jittered slightly for visualization purposes. LEFT: Left-skewed simulation, dark blue line, $T_{\text{opt}} = 21$, $a = 0.3$, $c = 2$. RIGHT: Right-skewed simulation, light blue line, $T_{\text{opt}} = 14$, $a = -0.3$, $c = 2$

2.3 Bayesian inference for LINEX TPC using `bayesTPC`

In R and most other common statistical software, the sort of extensions to GLMs used here are not easy to implement by default. Typically a user would need to be able to manually specify the predictor, link, and likelihood. This must then be inputted into an appropriate optimization routine (to perform maximum likelihood estimation of parameters), or priors could additionally be specified and Bayesian estimation performed (e.g., via Markov-chain Monte Carlo). This typically requires some specialist knowledge. However, the package `bayesTPC` (Sorek et al. 2025) in R (R Core Team 2025) allows for specification of TPCs and model formulations in a relatively straightforward way. Thus we use this package and its machinery to build a straightforward example pipeline.

Building a model in `bayesTPC` requires three steps:

1) defining a named TPC with corresponding formula (as an R "expression")
2) defining priors for all parameters that will be fitted as a vector with named elements
3) building the model using the appropriate model specification function – here `specify_binomial_model` which assumes the appropriate logit link function and binomial likelihood.

```
library("nimble")
library("bayesTPC")
```

```
my_name<- "LinExp_binom"
LinExp_formula <- expression(-exp( -a*(Topt-Temp) ) + a*(Temp-Topt) + cc)
LinExp_priors <- c( cc = "dexp(0.1)",
                    a = "dnorm(0, 1)",
                    Topt  = "dnorm(25, 1/1000)"
                    )
LinExp_binom <- specify_binomial_model( my_name,
                                        parameters = LinExp_priors,
                                        formula = LinExp_formula
                                        )
```

```
Model type 'LinExp_binom' can now be accessed using other bayesTPC functions. Use `reset_models()` to re
```

These functions build the model in the BUGS language. This resulting model can be examined here:

```
cat(configure_model(LinExp_binom))
```

```
{
    for (i in 1:N){
        logit(m[i]) <- ( -exp(-a * (Topt - Temp[i])) + a * (Temp[i] - Topt) + cc )
        Trait[i] ~ dbinom(m[i], n[i])
    }
    cc ~ dexp(0.1)
    a ~ dnorm(0, 1)
    Topt ~ dnorm(25, 1/1000)
}
```

Notice that the TPC expression is defined in terms of the temperature, which is notated as `Temp`. This is required by `bayesTPC` to ensure that all follow-on functions work properly. Parameters can otherwise be defined as the user prefers, although it is recommended that parameters not share names with functions or variables that are used elsewhere in R (for example `c` which is the concatenate function in R, or `T` and `F` which correspond to `TRUE` and `FALSE`).

3 Results

3.1 Performance on simulated data

The `bayesTPC` package requires that data be formatted as a list with elements called "Trait" (the response being modeled) and "Temp" (the corresponding temperature at which the trait is measured. Additionally, for fitting binomial models, as we are doing here, the data list should include "n", the number of trials of which the "Trait" is the number of successes in that trial.

```
sim.linexp.datl<-list(Trait = datl$counts,
                      Temp = datl$Temp,
                      n = datl$n)
sim.linexp.datr<-list(Trait = datr$counts,
                       Temp = datr$Temp,
                       n = datr$n)
```

Once the data have been formatted and our bespoke LINEX TPC model defined, we use the built in function `b_TPC` to fit the model. There are various options that can be included. Here we specify a sampler type "AF_slice" (that works well for posteriors that may have high correlation, as TPCs often do), as well as a burn in length and number of total iterations of the MCMC to keep (for more details about possibilities see CITE). We also give an example of specifying a starting value for the sampler. Although this is not strictly necessary, it can speed up convergence.

```
sim.linexp.fitl<-b_TPC(data = sim.linexp.datl,
                       model = "LinExp_binom",
                       samplerType = "AF_slice",
                       inits = list(a=0.1, Topt=17),
                       burn = 10000,
                       niter = 20000)


sim.linexp.fitr<-b_TPC(data = sim.linexp.datr,
                        model = "LinExp_binom",
                        inits = list(a=-0.1, Topt=15),
                        burn = 10000,
                        niter = 20000)
```

The `bayesTPC` package includes a variety of visualization tools that utilize `coda` (Plummer et al. 2006) and similar packages to allow checking of MCMC chain convergence, prior/posterior overlap, and marginal and joint posterior distributions of parameters (see Section A.1 for diagnostic plots for the model fits for the left-skewed data). It also provides summaries of model fits, and maximum *a posteriori* (MAP) estimates of parameters. For our purposes here, the most interesting output is the estimated median posterior TPC and the 95% highest posterior density (HPD) interval of this median across temperatures (Figure 3). In the simulated data set, we have a good amount of data across a reasonably fine grid of temperature settings. This allows the algorithm to reconstruct the curve and parameters well.

It is important to note that as the model is so flexible, it can miss the mark in cases where the experimental design (i.e., the temperature settings) is less than ideal. In Figure 3 (bottom two panels) we give examples of fits obtained when the data at a single temperature (here 10°C) are removed. For the left skewed data (Figure 3, bottom left panel), this removes data in the longer tail, which can result in two effects. First (as shown here) we can get a similar fit, with slightly increased uncertainty. Having this data missing can also slow convergence and result in poor fits where the estimated skew is backwards and the peak is close

to the range of the missing data (not shown). In contrast, when the data are missing closer to the peak as in the right-skewed data (Figure 3, bottom right panel) the right hand tail is still captured adequately, but the uncertainty in the location of the peak increases, which also changes the estimation of the temperature where the drop-off occurs. These are standard issues in estimation of parameters for non-linear functions if insufficient predictor settings are used, as has been noted by other authors (CITE SAMRAAT's PAPER).

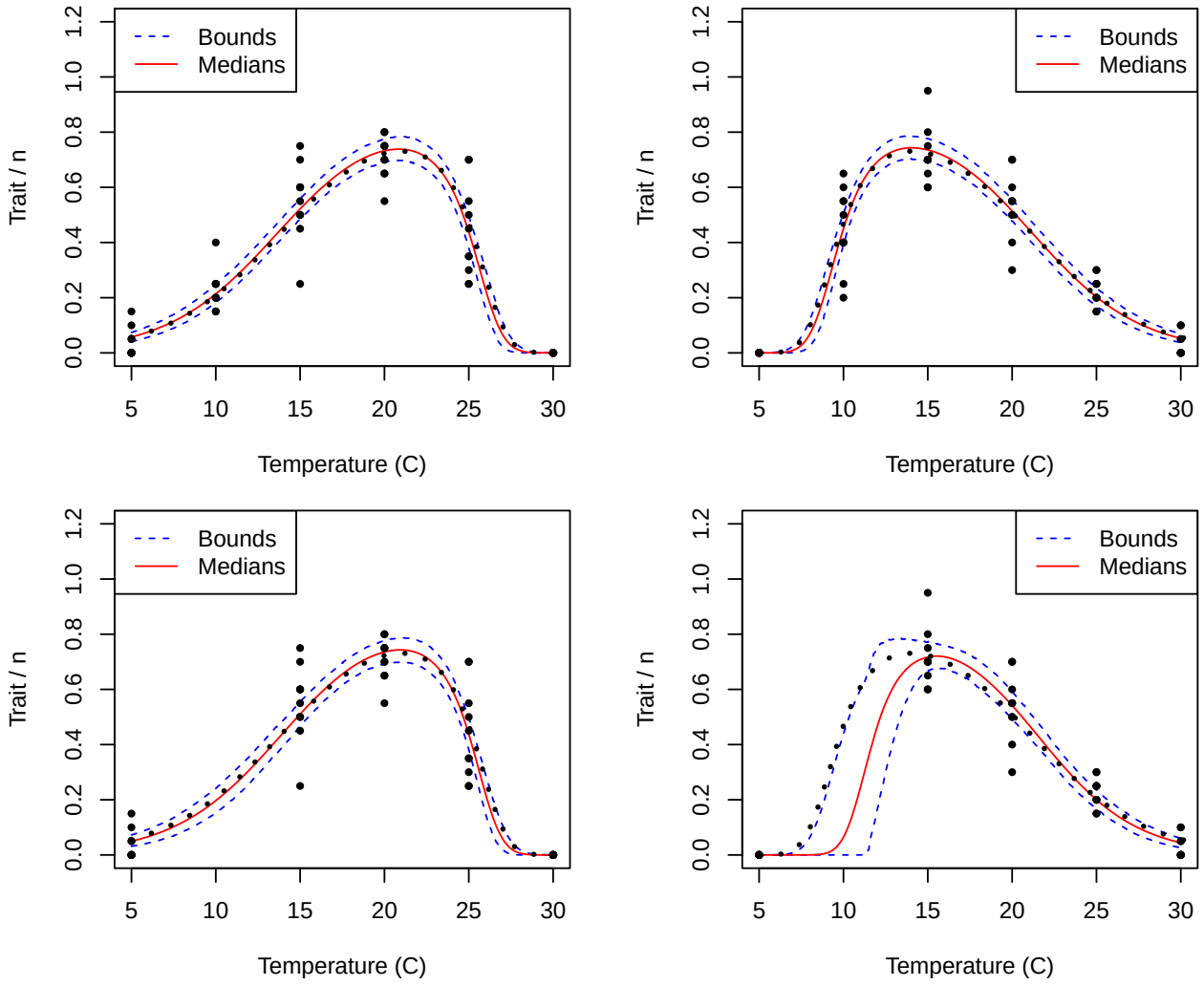


Figure 3: Visualization of posterior fits of the LINEX TPC to simulated data from the LINEX TPC model from the left-skewed (LEFT) and right-skewed (RIGHT) models show in Figure 2. The TOP row shows the complete data set and the BOTTOM row where the data at 10C has been removed. The median fit is indicated with a solid red line, the 95% HPD interval of the TPC as the dashed blue lines. The true line is indicated with a dotted black linke. Points correspond to 10 virtual replicates, each consisting of $n = 20$ 'trials' with the proportion of successes plotted (counts/n) at each temperature.

3.2 Examples on arthropod TPC data

Now we show the performance of the LINEX TPC on experimental data focusing on model selection. In many cases theory can guide the choice between a symmetric or asymmetric TPC. The LINEX function proposed here cannot capture symmetric patterns, so the user must consider assumptions and compare carefully. For example, compound traits, such as the probability of emergence of an adult mosquito (which integrates over multiple life stages), are often assumed to have symmetric TPCs. However, for traits for which the choice is less clear cut, or where secondary stressors may impact the trait, one will typically explicitly choose between functional forms. Thus we give two examples here – one where the default assumption would be symmetric and the other which is less clear cut – where we show how the user could fit multiple functional forms and choose between them using information criteria, specifically WAIC (widely-applicable information criterion).

3.2.1 Egg to adult emergence probability for *Anopheles stephensi*

For our first example, we examine data from Huxley et al. (2025) on the proportion of *Anopheles stephensi* eggs that successfully develop into adults across 9 temperatures and 3 different relative humidity (RH) levels. This trait aggregates survival and development across multiple life stages (egg, 3 instars, and pupa) into a single metric. TPCs for this trait are typically assumed to be symmetric (e.g., quadratic). We fit the LINEX TPC in the binomial likelihood framework as well as a more standard quadratic GLM to these data. We use WAIC as our metric for model selection.

RH	LINEX	QUAD
45	368.42	275.39
75	285.62	296.98
90	462.46	639.60

Table 1: WAIC values for the LINEX and quandraic generalized linear model fits from Figure 4.

We can see in Figure 4 that both models visually have reasonable fits at each humidity level, although the relatively flat emergence probability across a wide temperature range seen at 90% RH (bottom two rows) is

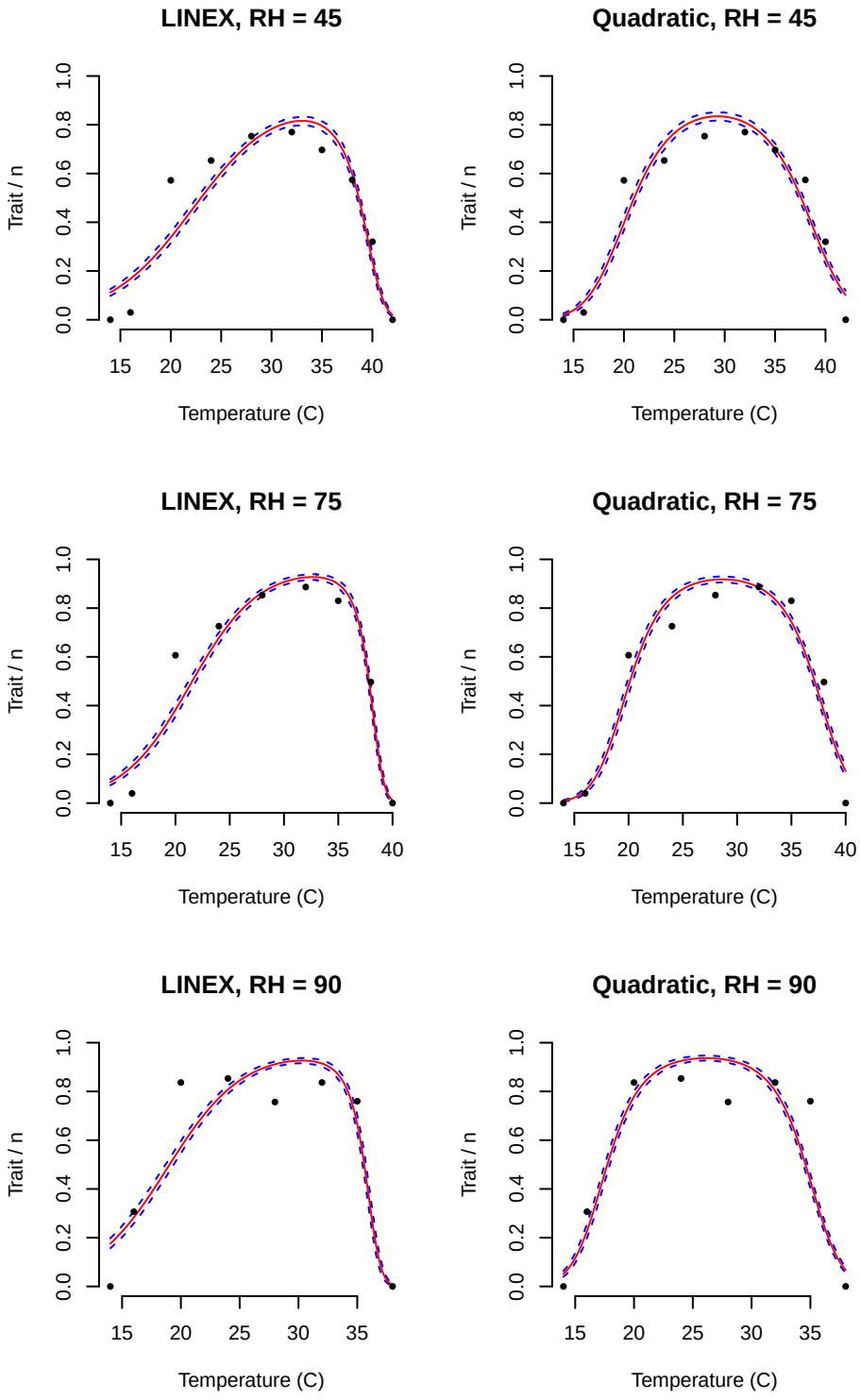


Figure 4: Visualization of posterior fits of the LINEX TPC and quadratic generalized linear model to data on the proportion of eggs emerging as adults across temperature for three humidity levels (from Huxley et. al, submitted). The median fit is indicated with a solid red line, the 95% HPD interval of the TPC as the dashed blue lines.

not particularly well captured by either model. Generally, model selection via WAIC prefers the symmetric quadratic function at most RH (including others not shown here), although at 75% RH the LINEX is favored here (Table 1).

3.2.2 Another set of trait data

Here we will put another set of the trait data, and analysis, focusing on one that is more likely to be asymmetric! Yay!

4 Discussion

In this paper we introduce the use of the LINEX function, a fairly simple non-linear function that can serve as an asymmetric TPC in a GLM-style modeling approach. This approach can be used model bounded data that are not well captured by a more classic piece-wise continuous bounded TPC with normal (or approximately normal) noise. We showed how to implement analyses using the `bayesTPC` package to perform inference in a Bayesian context, exploring both the performance on simulated data (where parameters are known) and experimental data, where parameters are unknown.

There are a wide variety of functions that have been proposed as TPCs in thermal ecology. These functions allow users to assume a wide variety of shapes while assuming different levels of "mechanism" that cause the shape. TPC function also vary in their flexibility. The Briere function (Briere et al. 1999) is fairly strict, with a shape that is always left skewed (long left-hand tail), a sharp drop off, and an assumption that minimum performance must occur at temperatures above 0°C. In contrast, the new flexTPC (Cruz-Loya, Mordecai, and Savage 2024) enables symmetric or non-symmetric shapes, left- or right-skew, and minimum temperatures of any value. The trade-off is an increase in the number of parameters needed to capture the possible shapes.

A unifying feature of these TPCs that they are explicitly bounded from below, that is the functions are ≥ 0. This boundedness is important for the vast majority of TPC applications, as the performance traits being modeled are also typically strictly ≥ 0. However, this makes established TPCs inappropriate for including within a GLM-type analysis with canonical link functions. Although it is strictly possible to include the more traditional (typically piece-wise continuous) TPCs in a GLM analysis using a unit link, this can create other problems. For example, neither the Poisson distribution nor the binomial distribution is well defined for a mean with value exactly zero. The canonical link functions explicitly force an assymptote *towards* zero to avoid this occurring. If one is instead truncating a function (e.g., a Briere) at particular values, this causes instabilities in estimation under this framework. One could instead use a small non-zero value outside of the minimum and maximum limits of the TPC. However, doing so is equivalent to assuming a minimum rate (for Poisson) or probability (for a binomial) that is constant across all other values. This is both a biologically

poor assumption, and one that can bias estimation of the TPC and predictions from the model.

The LINEX TPC, together with a simple quadratic (to capture symmetric responses), are able to represent a wide range of shapes that are appropriate to use within the logistic model framework because they have infinite support – that is they can produce responses from $-\infty$ to ∞. Further, because of this property, they are also appropriate to model other sorts of data within the GLM-flavor framework. For example, count data (such as may be recorded for fecundity) could be modeled with the LINEX TPC and the log-link to either a Poisson or negative-binomial likelihood. Thus, this relatively simple three-parameter function can be used for a wide variety of thermal trait (or other) data.

Although the model is fairly simple, due to the non-linearity in the parameters, the choice of priors for Bayesian inference can still be a bit of an art. The temptation is often to choose priors that are wide and relatively un-informative for all parameters, but this can result in inadequate fitting and bias. For example, if one lets both c and a have wide priors, particularly if both can be either positive or negative, the resulting fit can be biased. The rate of convergence can be greatly improved by inestad choosing weakly informative priors for one or both of these parameters. For instance, if we know that our peak probability is high-ish – above 0.3 or so – then a prior that constrains c to be positive improves the inference substantially. Further, if one has biological knowledge about the likely direction of the skew of the trait, constraining a to reflect this will also improve inference. In the examples above we chose weakly informative priors to substantially improve convergence. Practitioners taking a maximum likelihood approach would likely need to constrain optimization routines in similar ways to obtain good results.

5 References

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A Appendix

A.1 Diagnostics and summary for fit of left-skewed simulation example

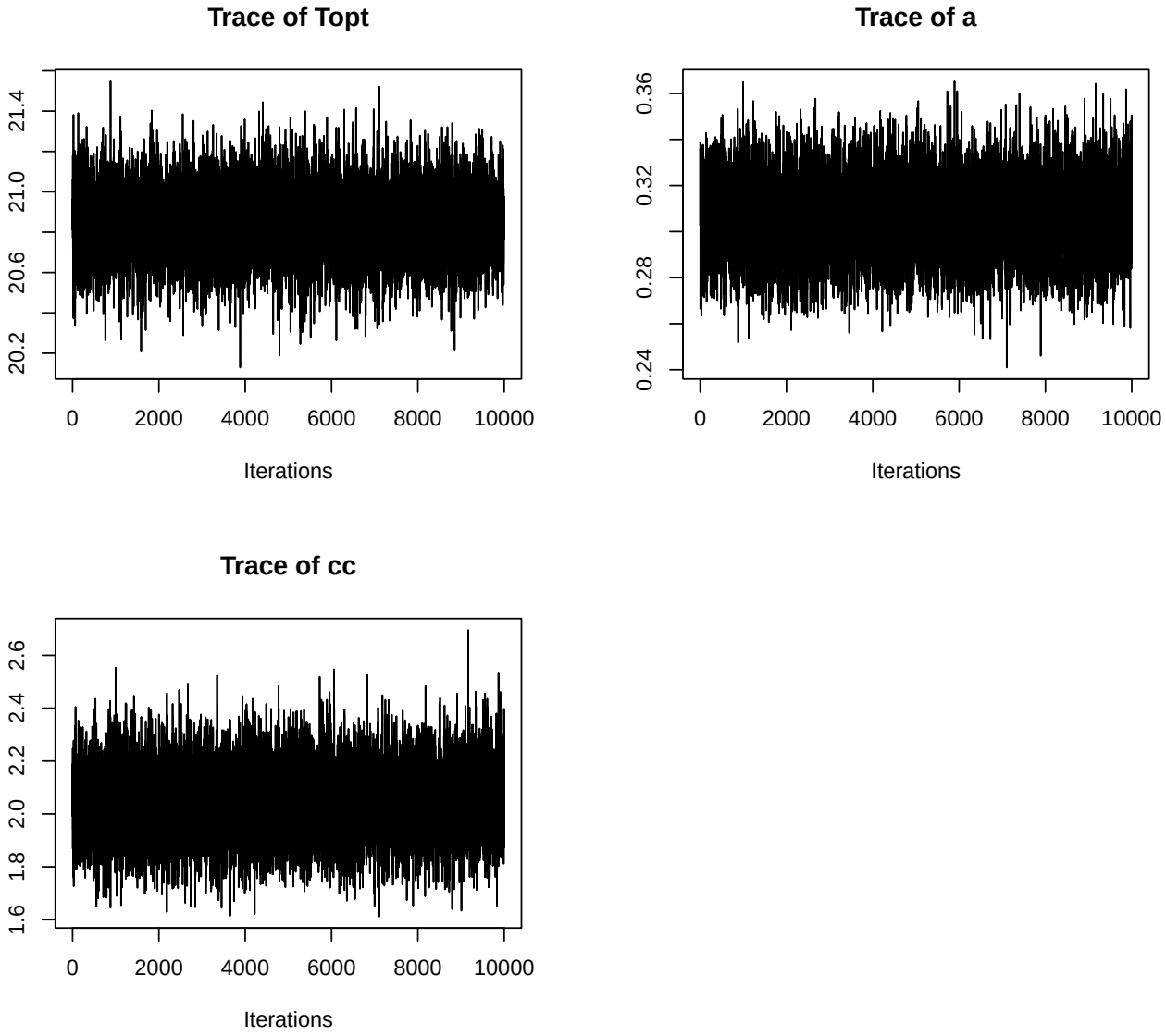


Figure A1: Trace plots for the LINEXP model fit to the left skewed simulation data.

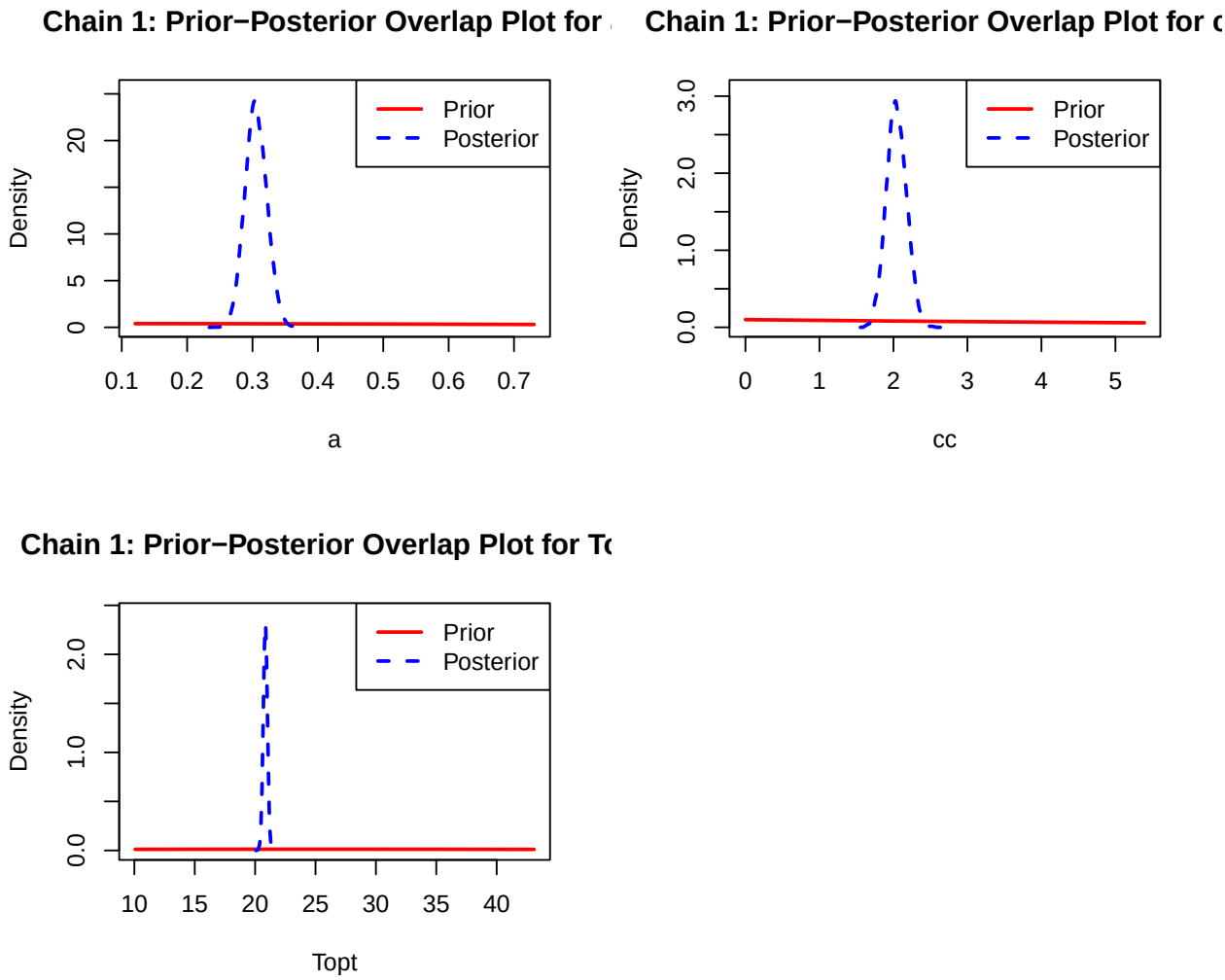


Figure A2: Prior-posterior overlap plots for the LINEXP model fit to the left skewed simulation data.

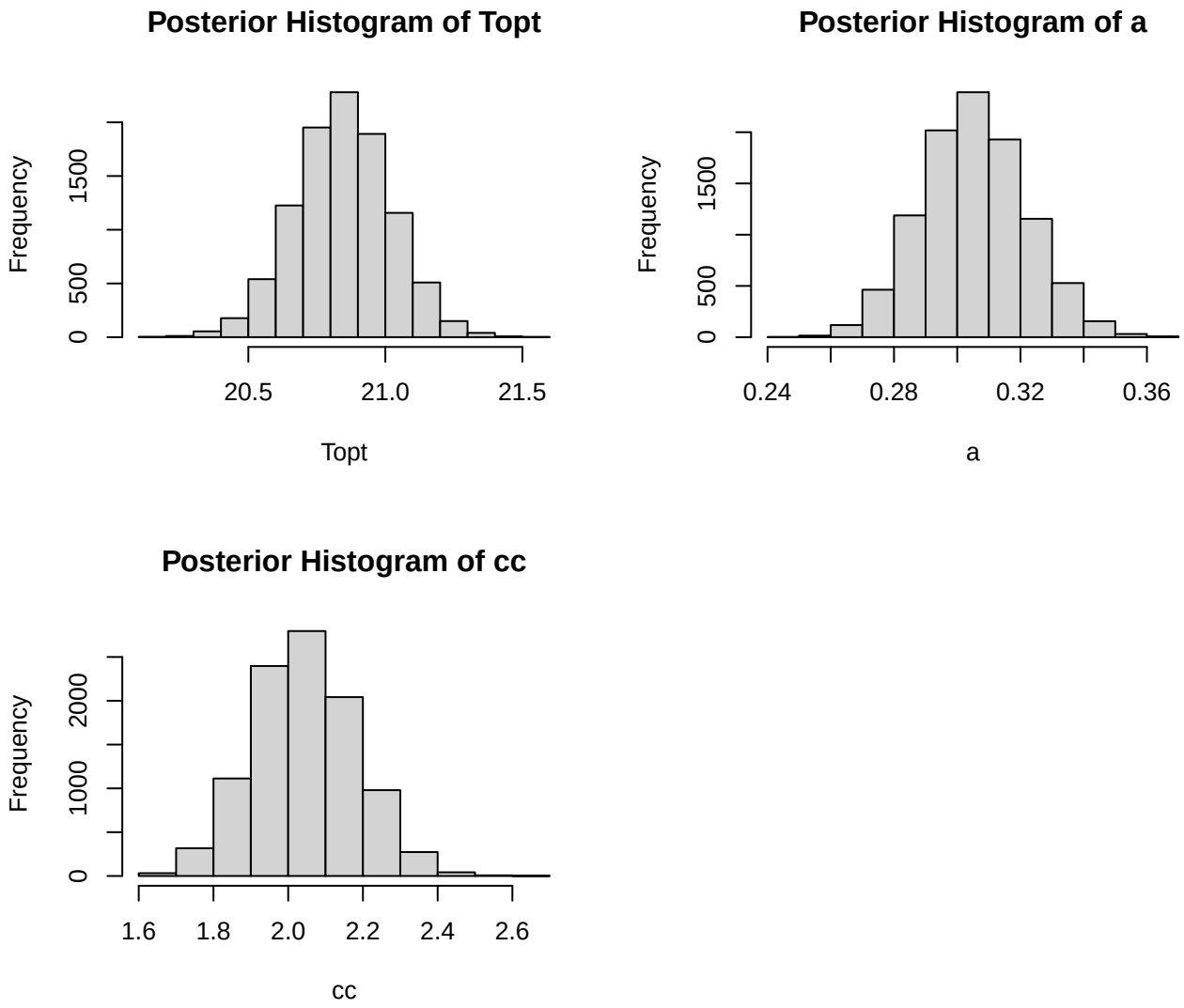


Figure A3: Histograms of the marginal posterior samples of parameters for the LINEXP model fit to the left skewed simulation data.

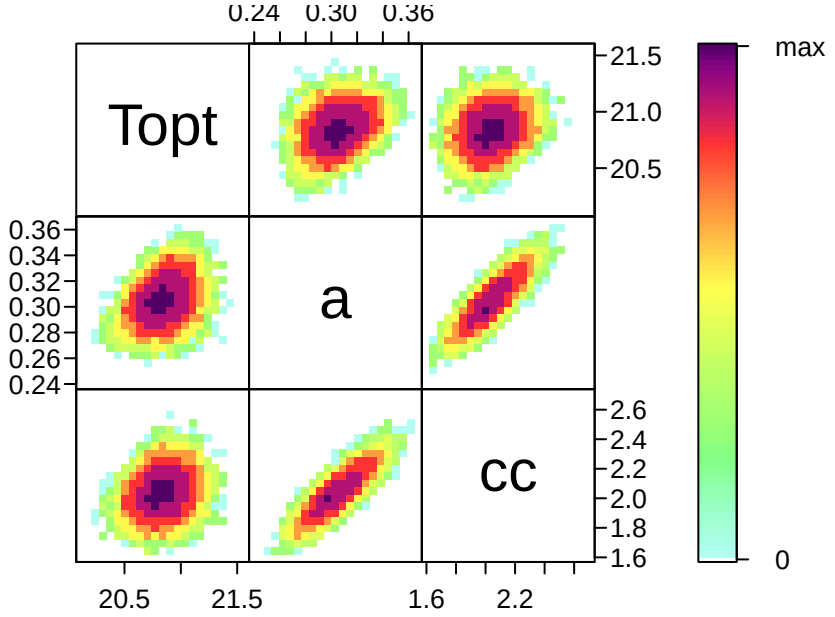


Figure A4: Pairs plot of samples from the joint posterior of parameters for the LINEXP model fit to the left skewed simulation data.

```r
summary(sim.linexp.fitl)
```

```
bayesTPC MCMC of Type:
  LinExp_binom

Formula:
  logit(m[i]) <- ( -exp(-a * (Topt - Temp)) + a * (Temp - Topt) + cc )

Distribution:
  Trait[i] ~ dbinom(m[i], n[i])

Priors:
  cc ~ dexp(0.1)
  a ~ dnorm(0, 1)
  Topt ~ dnorm(25, 1/1000)

Max. A Post. Parameters:
     Topt         a        cc  log_prob
  20.8346    0.3053    2.0439 -110.2282

MCMC Results:
Iterations = 1:10000
Thinning interval = 1
Number of chains = 1
Sample size per chain = 10000

1. Empirical mean and standard deviation for each variable,
```

```
   plus standard error of the mean:

          Mean      SD  Naive SE Time-series SE
Topt 20.8450 0.17487 0.0017487      0.0018069
a     0.3053 0.01649 0.0001649      0.0001755
cc    2.0429 0.13682 0.0013682      0.0014181

2. Quantiles for each variable:

          2.5%    25%     50%     75%   97.5%
Topt 20.5009 20.729 20.8465 20.9626 21.1831
a     0.2739  0.294  0.3049  0.3163  0.3382
cc    1.7776  1.951  2.0383  2.1338  2.3168
```